



# Phytochemical and acaricidal analysis of lavender (*Lavandula angustifolia*) and sage (*Salvia officinalis*) essential oils against *Rhipicephalus microplus* ticks: An in vitro and in vivo study

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## ABSTRACT

Based on earlier reports on acaricidal and repellent properties of *Lavandula angustifolia* and *Salvia officinalis* plants against various ectoparasites, including tick species, the present study was carried out using these essential oils (EOs) to explore the possibility of developing a potent herbal anti-tick formulation to manage cattle tick *Rhipicephalus microplus*. The EOs were evaluated against different life stages of ticks to provide a complete solution for tick control. The oils were also subjected to GC-MS analysis for their phytochemical constituents, which revealed a total of 125 compounds in lavender oil, predominantly consisting of alpha-pinene (10.24 %), D-limonene (7.50 %), 1,3,6-octatriene, 3,7-dimethyl-(Z) (6.48 %) and linalool (6.43 %), while sage oil contained 92 compounds, with linalool and linalyl acetate with an area percentage of 11.03 % and 9.83 % being the most abundant. All the treated engorged ticks were killed at 5 and 10 % EOs of *L. angustifolia* and *S. officinalis* with a significant reduction in oviposition in different concentrations. Both EOs were also effective against 83–100 % larvae at a 5 % dose in larval immersion test. Additionally, the study also examined the hatching inhibition of *R. microplus* eggs treated with both essential oils and achieved 100 % hatching inhibition at 5 % concentration. The repellency index (RI) indicated effective repellent activity with *L. angustifolia* having  $0.0 \pm 0.0$  and *S. officinalis*  $0.6 \pm 0.15$  at RI 10 %. Hematological and biochemical parameters showed no significant changes and no skin irritation in topically treated animals, suggesting the oils' safety for livestock use. The in vivo efficacy of both EOs was over 80 % after 72 h under patch treatment of naturally infested cattle. This research highlights the potential of lavender and sage essential oils as viable alternatives to develop an eco-friendly solution for tick management.

## 1. Introduction

*Rhipicephalus microplus* is an ectoparasite responsible for causing great economic losses to livestock industries worldwide, especially in tropical and subtropical regions. Massive tick infestation causes adverse impacts on the animals by creating various types of animal health conditions directly, such as anemia, loss of weight gain, tick worry, lower milk production, hide damage, and indirectly, it transmits pathogens of lethal diseases like anaplasmosis and babesiosis in livestock (Miraballes and Riet-Correa, 2018). Besides mosquitoes, ticks are globally considered as the most serious vectors of pathogens in animals (De la Fuente

et al., 2008). In India, the annual cumulative cost of tick control in cattle was predicted as 37.13–155.97 million USD, total annual loss due to hide damage was 92.10 million USD, and total annual economic milk loss due to tick-borne diseases (TBDs) 91.75 million USD (Singh et al., 2022).

Ticks also affect cattle productivity by inhibiting body weight gain by animals and thereby reduction in milk production. These factors have a significant impact on public health enforcing the essentiality of new anti-tick products to repel or control the larvae and adults of different ticks. In this direction, few methods were exploited for managing tick problems in which application of chemical acaricides has been widely

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used for 4–5 decades. Presently, Indian livestock owners prefer repeated spray of these chemicals mainly belonging to three major classes of insecticide i.e. carbamates, organophosphates and pyrethroids. All these have shown high tick control capacity since the beginning and managed tick infestations in animals. But recently, these chemicals have been reported as a serious threat to the environment, animals and human health (Staffa et al., 2020). In addition to environmental pollution, the resistance problem in ticks against different chemical acaricides has emerged in India and abroad. Accordingly, several reports have been published by many tick researchers from many regions of India (Sharma et al., 2012; Ghosh et al., 2015; Kumar et al., 2016; Sagar et al., 2020; Bisht et al., 2021; Fular et al., 2021; Khating et al., 2024). In addition, acaricidal resistance has also become a global problem of livestock across the world (Guerrero et al., 2012; Busch et al., 2014; Encinosa Guzman et al., 2016; Klafke et al., 2017) which causes hurdles in the successful accomplishment of integrated tick management program in a country (Michael et al., 2001). Acaricidal resistance in *R. microplus* developed mainly due to repeated and improper use of acaricides, which exerts selective pressure on tick populations, permitting only resistant ticks to survive (Kumar et al., 2020). Over time, it results in the reduction of efficacy of these chemicals.

To combat the problem of resistant tick populations, essential oils (EOs) offer a potential alternative due to their multiple modes of action, making it more difficult for ticks to develop resistance. In addition, EOs are biodegradable, environmentally friendly, and pose less risk of residual toxicity in animal products, proving them a sustainable option for integrated tick management (Salman et al., 2020). They consist of complex combinations of various phyto-compounds prepared from steam distillation, hydro-distillation, super-critical fluid extraction, or solvent-guided extraction process (Nakatsu et al., 2000). Essential oils may comprise many distinct plant metabolites structurally of different chemical classes in which terpenoids and phenyl propanoids are the main constituents in most of the essential oils (Carson and Hammer, 2011). Researchers from different laboratories of the world attempted to identify immune-modulatory, anti-microbial, anti-cancer, anti-inflammatory, anti-viral, anti-oxidant, and many other properties in various EOs (Bakkali et al., 2008). The EOs and terpenoids have shown bioactivity against various life stages of multiple tick species (Ribeiro et al., 2010). Various reports have published reproduction inhibitory action in tick species like inhibition in oviposition and egg hatching (Cetin et al., 2009; Abbas et al., 2014; Djebir et al., 2019). The variations in composition and inter-relationship among different active principles in EOs significantly influence acaricidal activity. Different bioactivities of EOs may be an effect of synergistic interaction of minor and major phyto-compounds (Castro et al., 2018). Furthermore, plant-derived substances have been characterized as cheap, minimal residual effects, and a reduced likelihood of inducing resistance (Rosado-Aguilar et al., 2010).

Plant-derived compounds may also accelerate herbal drug development process for tick control and have also been explored in the last decade against tick stages with satisfactory results (Mehlhorn et al., 2005; Mehlhorn, 2008; Semmler et al., 2011; Singh et al., 2014). Earlier, acaricidal and oviposition inhibitory activities of *Egletes viscosa* and *Lippia schaueriana* essential oils on *R. sanguineus* sensu lato engorged females were documented by Pereira et al. (2023). Several natural acaricides including plant oils, extracts and plant metabolites have been explored against various tick species showing their reproductive and hatching inhibitory and cidal activity (Oliveira et al., 2018). Lavender EO has been reported for its acaricidal and repellent effects against various tick species, including *Rhipicephalus (Boophilus) annulatus* and *Hyalomma marginatum rufipes* (Mkolo and Magano, 2007; Pirali-Kheirabadi and da Silva, 2010). Besides, it has been shown to reduce egg-laying in ticks, indicating its potential to disrupt the reproductive physiology of these ectoparasites (Djebir et al., 2019). Sage contains compounds such as camphor,  $\alpha$ -thujone, and 1,8-cineole, which have acaricidal activity against various ectoparasites, including ticks

(Nyabayo et al., 2015; Singh et al., 2022). Keeping in view, the present study was carried out utilizing EOs of lavender (*Lavandula angustifolia*) and sage (*Salvia officinalis*) against *R. microplus* following in vitro and in vivo approaches with their phytochemical analysis to promote eco-friendly management of resistant ticks.

## 2. Materials and methods

### 2.1. Procurement of essential oils (EOs)

The essential oils of *L. angustifolia* and *S. officinalis* of therapeutic grades were fetched from Biosur Pharma (Nagar Rajakhedi, Maharashtra) and were used to conduct research work.

### 2.2. Phytochemical analysis of essential oils

The phytochemical investigation of essential oils was conducted using GC–MS employing a GCMS–QP2010 Ultra Shimadzu Japan mass spectrometer, equipped with Electron Ionization (EI, 70 eV). Analytes were separated using the Rxi-5SII MS capillary column (30 m x 0.25 mm I.D., 0.25  $\mu$ m film thickness; Restek, USA) with a split ratio of 1:10 and an injection volume of 1  $\mu$ L. The gas chromatography oven temperature protocol started at 40°C (maintained for 5 min), and thereafter increased at a rate of 20°C per minute to 280°C (held for 8 min), resulting in a total cycle duration of 25 min. Helium (99.999 %; Li ming Gas Co., Harbin, China) was used as a carrier gas at a constant flow rate of 1.2 mL per min. The source temperature was maintained at 220°C, whilst the interface temperature was regulated at 300°C. The retention time of the targeted metabolites was measured in full scan mode by capturing the mass range from  $m/z$  40–600. Quantitative data was obtained using selective reaction monitoring (SRM) mode. Details on all precursor and product ions, along with the collision energy used, are included in the ESI.

### 2.3. Tick sampling and processing

Freshly dropped adult female ticks were collected from animal sheds of different households and dairy farms situated in different blocks of Indore district of Madhya Pradesh state. It is located in the western region of state at 76°E and 23°N, on the southern edge of the Malwa plateau with an average elevation of 553 m above sea level on the Saraswati and Khan rivers, which are tributaries of the Shipra River. The ticks collected from each block following random sampling procedure were pooled to form one sample, designated as one population. Engorged females of *R. microplus* were gathered from the ground, as well as from the cracks and crevices found in the animal sheds. Following the preliminary processing of gathered samples at the collection stations, the ticks were subsequently transported to the Department of Medicine, College of Veterinary Science and Animal Husbandry, Mhow (NDVSU), Indore (Madhya Pradesh). The collected ticks were washed and maintained temperature at 28°C and 85  $\pm$  5 % relative humidity (RH) in the Biological Oxygen Demand (BOD) incubator for conducting adult-based assay (Ghosh et al., 2015). Few adult females were pooled together block-wise and kept in glass tubes for egg laying. The larvae hatched from these eggs were used in larval bioassays.

### 2.4. Identification of ticks

Collected ticks were morphologically identified as per the keys described by Walker et al. (2003). Before performing the bioassays, the collected engorged females of *R. microplus* were examined under a zoom stereoscopic microscope. The females ticks were identified based on morphological characters like short mouth parts, possession of hexagonal basis capitulum, bifid first coxae, longer than broad scutum, and absence of anal groove.

## 2.5. Preparation of test concentrations of essential oils

Sequential concentrations of 0.625, 1.25, 2.5, 5, and 10 % from each essential oil were formulated in 10 % absolute ethanol and 2 % Triton X-100 for execution of Egg Hatch Assay (EHA), Larval Immersion Test (LIT), Adult Immersion Test (AIT), Larval Packet Test (LPT), and Repellency Index (RI) bioassays. The same concentration range was chosen for various tests and tick stages to ensure uniformity across different assessments, hence enabling a direct comparison of the effectiveness of essential oils across life stages and testing methodologies.

## 2.6. Bioassays for in vitro study

### 2.6.1. Adult immersion test (AIT)

The assay was performed following the protocols established by Sharma et al. (2012), with slight adjustments made to the original methods. The evaluation of essential oils was performed using 0.625, 1.25, 2.5, 5 and 10 % concentrations as mentioned above in Section 2.5. Each concentration of essential oil was replicated three times, and 10 pre-weighed freshly dropped engorged *R. microplus* ( $n = 3 \times 10$ ) were immersed for 10 min in each concentration. After treatment, the ticks were removed and placed on tissue paper before being transferred into the Petri dishes. The control group of ticks received treatment with the solvent. Petri dishes were stored in desiccators located within a BOD incubator, which was maintained at a temperature of 28°C and a relative humidity of  $85 \pm 5$  %. Every 24 h, each Petri dish was scrutinized to assess mortality over a period over up to 14 days. Engorged females who failed to lay eggs within 14 days were classified as deceased. The eggs produced by the female ticks were carefully weighed and placed into glass vials, which were then covered with cotton fabric and secured with an elastic band. The entomological parameters and overall efficacy of essential oils were assessed (Bennett, 1974; Ghosh et al., 2013) as follows:

- i. Reproductive index (RI) = 
$$\frac{\text{Weight of eggs laid by female (mg)}}{\text{Weight of Female before egg laying (mg)}}$$
- ii. Percentage inhibition of oviposition (%IO) = 
$$\frac{\text{RI}_{\text{control}} - \text{RI}_{\text{treated}}}{\text{RI}_{\text{control}}} \times 100$$
- iii. Overall efficacy (E%) =  $100 [1 - (\text{CRT} \times \text{CRO})]$

Where, CRT = Mean number of dead ticks in treatment / Mean number of live ticks in control; CRO = Mean egg mass in treatment / Mean egg mass in control

### 2.6.2. Larval packet test (LPT)

The testing of EOs was carried out by LPT as described previously (FAO, 2004) with minor modifications. For this purpose, an amount of 0.6 mL of each concentration of essential oil (*L. angustifolia* and *S. officinalis*) prepared in 10 % ethanol and 2 % Triton X-100 in water and was used to impregnate the filter paper rectangles measuring 3.75 cm by 8.5 cm (541 Whatman, Maidstone, Kent, and United Kingdom). These impregnated filter papers were undergone a drying process in an incubator set at 37°C for a minimum duration of 30 min. The rectangles were folded in half and secured along the sides with adhesive tapes, resulting in the creation of an open-ended packet. Following the insertion of around 150 larvae (7–10 days old), each packet was securely sealed using “bulldog” clips, and the packets were then positioned in a BOD incubator set to 28°C with a relative humidity of  $85 \pm 5$  %. After 24 h, the packets were taken out, and both dead and live larvae were counted under an electric lamp, allowing for the estimation of larval mortality.

### 2.6.3. Larval immersion test (LIT)

The acaricidal activity of the essential oils was assessed on tick larvae through the Larval Immersion Test (LIT) as outlined by Ajithkumar et al. (2016). In the bioassay, over 100 larvae per replication were placed into

1.5 mL centrifuge tubes, followed by the addition of 200 µl of each concentration for the immersion of the larvae. Following a 10-minute exposure to the larvae, the test solution was extracted from the tube, and any excess solution was absorbed using a filter paper strip. The tubes were subsequently encased in cotton cloths secured with rubber bands. The control group was immersed in the solvent utilized for the preparation of the essential oil concentration. For each concentration, three replications were maintained, and treated larvae were placed separately in a BOD incubator at 28°C and  $85 \pm 5$  % RH conditions. Following 24 h, the tubes were opened, and both dead and live larvae were counted to determine the mortality rate.

### 2.6.4. Egg hatch assay

The egg hatch assay was performed following the methodology outlined by Haque et al. (2014), with a few minor modifications adopted. About 5 mg (equivalent to 125 eggs  $\pm 5$ ) of *R. microplus* were introduced into a glass tube and submerged in 1 mL of each concentration of essential oil for duration of 10 min (Singh et al., 2022). Following this, the solution was decanted, and after the solvent was absorbed by filter paper strips, the tubes were covered with muslin cloth. The experiments were conducted three times, including a distilled water control, and were incubated at 28°C with  $85 \pm 5$  % relative humidity for the duration of 14 days, until hatching was completed. The entomological data was recorded as follows (Haque et al., 2014):

- a) Hatching % =  $[\text{No. of hatched eggs} / \text{Total no. of eggs}] \times 100$
- b) Percentage inhibition of hatching (IH %) =  $[(\text{Hatching \% in control} - \text{Hatching \% in treated}) / \text{Hatching \% in control}] \times 100$ .

### 2.6.5. Evaluation of repellent activity

For adults

The repellent activity of EOs was assessed by choice test, according to methods (Bissinger et al., 2009; Ferreira et al., 2017). The EOs were diluted in distilled water with 2 % triton X-100 and 10 % ethanol to form a 10 % concentration for evaluation of their repellent activity. The test was performed on *R. microplus* adult females at room temperature ( $26 \pm 2^\circ\text{C}$ ). Filter papers were cut in half, in one half of the paper, 200 µL of the EO solutions were applied for each essential oils with a micropipette. In the other half, only diluent was applied. The treated filter papers were dried for 30 min at room temperature and then placed inside Petri dishes. Ten ticks were released at the center of each Petri dish, which was then placed in the absence of light for 30 min. After 30 min, the positions of ticks were assessed. The repellency index (RI) was calculated according to the method of Xavier et al. (2015) utilizing the following equation:

$$\text{RI} = 2T / (T + C)$$

Where, T = percentage of ticks attracted to the treatment and, C = percentage of ticks attracted to the control (solvent). RI values varied between 0 and 2, where  $\text{RI} < 1.0$  indicated a repellent effect and  $\text{RI} > 1.0$  indicated an attractive effect.  $\text{RI} = 1.0$  corresponded to a neutral effect.

For larvae

The repellency of tested EOs was evaluated by trapeze experiments. The filter paper was cut into strips of  $2 \times 15$  cm, and a line was inscribed across the center with a graphite pencil. Essential oils were dissolved in distilled water containing 2 % Triton X-100 and 10 % ethanol. A small volume of 100 µL of each dilution was applied to the center of the strip (at the pencil line). The control experiments were set using strips treated with diluents. The strips were let to dry for 20–30 min at ambient temperature. Approximately, sixty larvae were placed at the base of each strip using a paintbrush. The tick larvae traversing the center of the strip at 30 min were counted. Each EO was evaluated in triplicate concurrently. The experiment was conducted twice using freshly made dilutions of the EOs and new batches of larvae. The repellency index was determined as per the method of Xavier et al. (2015).

## 2.7. Skin irritancy test of EOs on animals

Prior to conducting in vivo trial, the cattle calves of 3–6 months old were randomly selected in two groups I and II having 5 calves in each (group I was treated with lavender EO and group II was treated with sage EO) for evaluation of skin irritancy test of lavender and sage EOs at 5 % for topical usage (Jain et al., 2021). The EOs were topically applied at the rate of 15 mL/patch in three patches measuring 6"x6" area per animal. Any visible changes such as erythma and pruritus at application sites were recorded before and 24 h post application. Besides, the hematological (Total leucocyte count, neutrophils, lymphocytes, monocytes, eosinophils, basophils, hemoglobin, total erythrocyte count, packed cell volume) and biochemical parameters (alanine transaminase, aspartate transaminase, creatinine) of animals were also measured after 24 h.

## 2.8. In vivo efficacy trial

The efficacy experiment was performed on naturally tick-infested cross-bred cattle under field conditions in or around Mhow as per the method of Sharma et al. (2022) with some modifications. A total of 12 crossbred cattle exhibiting an average of over 20 tick counts across various body areas were chosen for the study. The animals were randomly divided into two groups, each having six heads of cross-bred cattle. The first group received treatment with *L. angustifolia* EO at a concentration of 5 %, while the second group was treated with same concentration of *S. officinalis* EO. About 15–20 mL of the EO solution was applied by hand sprayer on selected tick-bearing patches of the animals. The concentration of the essential oils was determined by in vitro experiments. The in vivo efficacy of the essential oils was evaluated at 24, 48, and 72 h post-exposure. Anti-tick efficacy of each EO was estimated using the following equation (Holdsworth et al., 2006).

$$AE = [B - T/B] \times 100$$

Where, AE is the anti-tick efficacy, B is the mean number of surviving ticks before treatment, and T is the mean number of surviving ticks after treatment on a particular day.

## 2.9. Statistical analysis of data

All the experiments performed as per the statistical design. Number of observations were repeated and all the data were statistically analyzed using Graph Pad Prism version 5.00 (GraphPad Software, San Diego, California USA). Dose-response data obtained from LPT, LIT and AIT were analyzed by probit analysis using probit-transformed % mortality and log-transformed % concentrations to determine the LC<sub>50</sub> and LC<sub>95</sub> values of EOs (Finney, 1962). Significance of difference between mean values of entomological, hematological and biochemical data of treated and control group was determined by *t*-test. Statistical differences across treatments and concentrations were assessed using a one-way ANOVA, followed by Tukey's test for multiple mean comparisons (Snedecor and Cochran, 1994).

## 3. Results

### 3.1. Phytochemical analysis of EOs

The phytochemical analysis using GC-MS revealed 125 compounds in *L. angustifolia* essential oil. Chromatogram peak area (Area %) reflected the relative proportion of each compound. The retention time (in minutes) indicated the time taken by each compound to travel through the GC column before being detected. These compounds were separated based on their volatility and interaction with the column material. Alpha-pinene (10.24 %) was identified as a major compound in this analysis, followed by D-limonene (7.50 %), 1,3,6-octatriene (6.48 %), and linalool (6.43 %) in the EO of *L. angustifolia* (Table 1, listed only top 22 compounds on the basis of % area of distribution). However, GC-MS

**Table 1**

Phytochemical profile of *L. angustifolia* essential oil through GC-MS analysis.

| S. No | Name of compounds                                    | Retention time | Area % |
|-------|------------------------------------------------------|----------------|--------|
| 1.    | Tricyclo [2.2.1.0 (2,6)] heptane, 1,7,7-trimethyl    | 7.623          | 1.37   |
| 2.    | Tricyclo [2.2.1.0 (2,6)] heptane, 1,7,7-trimethyl    | 7.690          | 1.06   |
| 3.    | Alpha-Pinene                                         | 7.924          | 10.24  |
| 4.    | Camphene                                             | 8.128          | 5.75   |
| 5.    | Bicyclo[4.1.0]heptane, 3,7,7-trimethyl-, [1S-(1.alp  | 8.514          | 4.43   |
| 6.    | Beta-Myrcene                                         | 8.694          | 1.72   |
| 7.    | Alpha-Phellandrene                                   | 8.872          | 1.30   |
| 8.    | 1,3-Cyclohexadiene, 1-methyl-4-(1-methylethyl)       | 8.948          | 1.93   |
| 9.    | D-Limonene                                           | 9.210          | 7.50   |
| 10.   | 1,3,6-Octatriene, 3,7-dimethyl-, (Z)                 | 9.277          | 6.48   |
| 11.   | Gamma-Terpinene                                      | 9.523          | 1.36   |
| 12.   | Linalool                                             | 10.099         | 6.43   |
| 13.   | (+)-2-Bornanone                                      | 10.444         | 2.02   |
| 14.   | (+)-2-Bornanone                                      | 10.603         | 4.12   |
| 15.   | Bicyclo[2.2.1]heptan-2-ol, 1,7,7-trimethyl-, (1S-en  | 10.684         | 4.34   |
| 16.   | Bicyclo[2.2.1]heptan-2-one, 5,5,6-trimethyl-, exo    | 10.808         | 2.15   |
| 17.   | Alpha-Terpineol                                      | 10.924         | 2.22   |
| 18.   | Bicyclo[3.1.0]hexan-3-ol, 4-methyl-1-(1-methylethyl  | 10.972         | 1.41   |
| 19.   | Cyclohexanol, 1-methyl-4-(1-methylethylidene)        | 11.723         | 2.40   |
| 20.   | Caryophyllene                                        | 12.831         | 5.22   |
| 21.   | 1,4,7-, Cycloundecatriene, 1,5,9,9-tetramethyl-, Z,Z | 13.071         | 4.21   |
| 22.   | 2H-2,4a-Methanonaphthalen-8(5 H)-one, 1,3,4,6,7      | 14.293         | 1.63   |

analysis revealed 92 compounds in *S. officinalis* EO with linalool (7.23 %) followed by (R)-Methyl-cyclohexenol (5.74 %) and linalyl acetate (6.55 %) being the most abundant among the constituents (Table 2, listed only top 22 compounds). These compounds are primarily oxygenated monoterpenes and esters contributing to the aromatic and therapeutic activities of the EO. While the variations in the retention time indicated the order of these compounds elute during the analysis, correlating with their boiling points and polarity.

**Table 2**

Phytochemical profile of *s. officinalis* essential oil through GC-MS analysis.

| S. No | Name of compounds                                   | Retention time | Area % |
|-------|-----------------------------------------------------|----------------|--------|
| 1.    | beta. -Myrcene                                      | 8.685          | 3.05   |
| 2.    | Eucalyptol                                          | 9.210          | 3.43   |
| 3.    | 1,3,6-Octatriene, 3,7-dimethyl-, (Z)                | 9.375          | 1.51   |
| 4.    | Benzyl alcohol                                      | 9.563          | 2.80   |
| 5.    | Linalool                                            | 9.938          | 3.80   |
| 6.    | Linalool                                            | 10.153         | 7.23   |
| 7.    | 2,4,6-Octatriene, 2,6-dimethyl-, (E,Z)              | 10.253         | 1.60   |
| 8.    | 3-Cyclohexen-1-ol, 1-methyl-4-(1-methylethyl)       | 10.352         | 1.32   |
| 9.    | (+)-2-Bornanone                                     | 10.544         | 4.76   |
| 10.   | Bicyclo[2.2.1]heptan-2-ol, 1,7,7-trimethyl-, (1S-en | 10.646         | 1.59   |
| 11.   | (R)-Methyl-cyclohexenol                             | 10.825         | 5.74   |
| 12.   | Cyclohexanol, 1-methyl-4-(1-methylethylidene)       | 10.942         | 3.22   |
| 13.   | 2,6-Octadien-1-ol, 3,7-dimethyl-, (Z)               | 11.268         | 1.89   |
| 14.   | Linalyl acetate                                     | 11.323         | 3.28   |
| 15.   | Linalyl acetate                                     | 11.448         | 6.55   |
| 16.   | 2,6-Octadien-1-ol, 3,7-dimethyl-, acetate, (Z)      | 12.212         | 1.56   |
| 17.   | Geranyl acetate                                     | 12.355         | 2.65   |
| 18.   | Caryophyllene                                       | 12.803         | 3.89   |

### 3.2. Effect on adult females of *R. microplus*

Both the essential oils showed a dose-dependent effect on tick survival. The higher concentrations (5 % and 10 %) of both EOs provided nearly total mortality in treated ticks (100 %). Similarly, egg mass production was inhibited as the concentration increased, with complete reduction of egg production in 5 and 10 % EOs. This indicated that higher concentrations ceased complete reproduction in treated engorged females. Lower concentrations of both EOs caused 58.33–88.66 % reduction in oviposition egg-laying compared to the control. The overall efficacy of the EOs combining the data of mortality and % IO was observed from 82.2 % to 100 % in different concentrations of both EOs with no further reproduction and survival. The mortality slopes were determined to be  $2.869 \pm 0.57$  and  $2.508 \pm 0.58$ , respectively (Tables 3 and 4). The LC<sub>95</sub> values of *L. angustifolia* and *S. officinalis* EO were determined as 4.39 (3.70–5.20 % CI) and 8.03 % (6.58–9.78 % CI), respectively (Table 4).

### 3.3. Effect on *R. microplus* larvae in LPT

Different concentrations of *L. angustifolia* and *S. officinalis* EOs were exposed to the larvae as presented in Table 4 and the highest mortality of 83.77 and 81.55 % was recorded at 10 % concentration. The mortality slope for *L. angustifolia* and *S. officinalis* oil were calculated as  $1.691 \pm 0.086$  and  $1.753 \pm 0.136$ , respectively, and their LC<sub>50</sub> values were 2.45 and 2.80 %, respectively (Tables 4 and 5). Both oils were lethal against larvae of *R. microplus*. The coefficient of determination ( $R^2$ ) values were noted as 0.99 and 0.98, and compared.

### 3.4. Effect on *R. microplus* larvae in LIT

In larval immersion test, these EOs were found effective against *R. microplus* larvae, but *L. angustifolia* was comparatively more impressive as confirmed by lower LC<sub>50</sub> and LC<sub>95</sub> values showing 0.81 % (0.73–0.89 % CI) and 3.53 % (2.88–4.31 % CI), respectively. The steep mortality slopes and high  $R^2$  values showed consistent and expected mortality patterns with increasing concentrations. The highest larval mortality of 83–100 % was recorded at concentrations of 5 and 10 %, of both EOs. The mortality slopes were recorded as  $2.478 \pm 0.3955$  and  $2.336 \pm 0.3875$  (Tables 4 and 5).

### 3.5. Effect on egg hatching

Inhibition of hatching (IH %) and hatching percentage were performed to assess the effect of essential oils on eggs of *R. microplus*. The complete inhibition of hatching was recorded at 10 % concentration of

*L. angustifolia* EO and at 5 and 10 % of *S. officinalis* EOs (Table 6). It was noticed that hatching inhibition was influenced by the increasing concentrations of both EOs as compared to control where more than 97 % hatching was recorded.

### 3.6. Repellency activity

Both the essential oils used in the present study showed good repellent activity against adult females of *R. microplus* (Table 7). The repellency index (RI) values for the tested adult ticks were  $0.0 \pm 0.0$  and  $0.6 \pm 0.15$ , which were less than 1 showing good repellent activity. Against larvae of *R. microplus*, the repellent activity was clearly exhibited by EOs of both plants at different concentrations showing dose-dependent repellent effect from  $0.0 \pm 0.0$ – $0.64 \pm 0.16$  RI (Fig. 1). It is important to note that a strong repellency was noticed in 5 and 10 % *L. angustifolia* and 10 % *S. officinalis* essential oil (Table 8).

### 3.7. Safety study of EOs

These EOs were evaluated on animals for safety study and these treatments were found safe without showing any adverse effects on the cattle. We noted no skin irritation in animals, and the hematological and biochemical parameters obtained from the treated animals also showed no significant change ( $p < 0.05$ ) in the tested clinical parameters and were within the normal limits, indicating the clinical safety of these essential oils (Table 9).

### 3.8. In vivo anti-tick efficacy

The results of in vivo trials with *L. angustifolia* and *S. officinalis* essential oils (at 5 % concentration) in naturally infested cattle showed more than 84 % effectiveness in reducing the ticks over 72 h. A sequential reduction pattern in tick number on the animals was observed up to 72 h and the efficacy of *L. angustifolia* and *S. officinalis* EO was achieved as 88.64 and 84.68 %, respectively (Fig. 2 and Table 10). *L. angustifolia* was superior to *S. officinalis* exhibiting slightly better tick control. These results indicated anti-tick potential of EOs to develop natural acaricides to manage the tick problem.

## 4. Discussion

Ticks have recently drawn significant attention and are now regarded as the second most important vectors of human infectious illnesses globally, behind mosquitoes. Various tactics, such as biological and chemical management, have effectively diminished vector tick populations. The persistence of erratic use of synthetic acaricides enhanced

**Table 3**

Dose-response data of *Lavandula angustifolia* and *Salvia officinalis* against engorged female of *R. microplus*.

| Essential oils         | Concentration (%) | Number of ticks |      | Tick wt. (mg) (Mean $\pm$ SE) | Mortality (%) | Egg wt. (mg) (Mean $\pm$ SE) | Reproductive index | Inhibition of oviposition (%) | Overall Efficacy (%) |
|------------------------|-------------------|-----------------|------|-------------------------------|---------------|------------------------------|--------------------|-------------------------------|----------------------|
|                        |                   | Treated         | Dead |                               |               |                              |                    |                               |                      |
| <i>L. angustifolia</i> | 0.625             | 30              | 9    | 940 $\pm$ 32.14               | 30.00         | 51.66 $\pm$ 4.40             | 0.05               | 58.33                         | 87.6                 |
|                        | 1.25              | 30              | 13   | 890 $\pm$ 10                  | 43.33         | 40.0 $\pm$ 15.25             | 0.04               | 66.66                         | 86.1                 |
|                        | 2.5               | 30              | 20   | 956.66 $\pm$ 43               | 66.66         | 33.33 $\pm$ 12.01            | 0.03               | 75.00                         | 82.2                 |
|                        | 5                 | 30              | 30   | 1053.33 $\pm$ 66              | 100           | 0 $\pm$ 0                    | 0.00               | 100                           | 100.0                |
|                        | 10                | 30              | 30   | 843 $\pm$ 32                  | 100           | 0 $\pm$ 0                    | 0.00               | 100                           | 100.0                |
|                        | Control           | 30              | 0    | 966.66 $\pm$ 12               | 0.00          | 125.0 $\pm$ 2.88             | 0.12               | 0.00                          | -                    |
|                        | 0.625             | 30              | 7    | 1013.33 $\pm$ 46              | 23.33         | 45.0 $\pm$ 2.88              | 0.044              | 70.66                         | 93.4                 |
| <i>S. officinalis</i>  | 1.25              | 30              | 9    | 973.33 $\pm$ 3.3              | 30.00         | 31.66 $\pm$ 1.66             | 0.032              | 78.66                         | 94.0                 |
|                        | 2.5               | 30              | 15   | 970 $\pm$ 11                  | 50.00         | 17 $\pm$ 11.78               | 0.017              | 88.66                         | 94.6                 |
|                        | 5                 | 30              | 24   | 963.33 $\pm$ 49               | 73.33         | 2.5 $\pm$ 1.44               | 0.002              | 98.66                         | 98.7                 |
|                        | 10                | 30              | 30   | 1090 $\pm$ 46                 | 100           | 0 $\pm$ 0                    | 0.00               | 100                           | 100.0                |
|                        | Control           | 30              | 0    | 1040 $\pm$ 34                 | 0.00          | 158.33 $\pm$ 17.63           | 0.15               | 0.00                          | -                    |

**Table 4**Mortality slope,  $R^2$ ,  $LC_{50}$ , and  $LC_{95}$  values with 95 % CI of engorged *R. microplus* against *Lavandula angustifolia* and *Salvia officinalis* essential oils.

| Essential oils         | Bioassay | Slope $\pm$ SE   | $R^2$ | $LC_{50}$ (95 % CI) | $LC_{95}$ (95 % CI) | *P value |
|------------------------|----------|------------------|-------|---------------------|---------------------|----------|
| <i>L. angustifolia</i> | AIT      | 2.869 $\pm$ 0.57 | 0.89  | 1.17 (1.07–1.27)    | 4.39 (3.70–5.20)    | 0.0149   |
| <i>S. officinalis</i>  | AIT      | 2.508 $\pm$ 0.58 | 0.85  | 1.78 (1.51–2.09)    | 8.03(6.58–9.78)     | 0.0233   |
| <i>L. angustifolia</i> | LPT      | 1.691 $\pm$ 0.09 | 0.99  | 2.45 (2.12–2.82)    | 22.85(17.04–30.63)  | 0.0003   |
| <i>S. officinalis</i>  | LPT      | 1.753 $\pm$ 0.14 | 0.98  | 2.80 (2.44–3.20)    | 24.15(18.21–32.02)  | 0.0010   |
| <i>L. angustifolia</i> | LIT      | 2.478 $\pm$ 0.39 | 0.92  | 0.81 (0.73–0.89)    | 3.53(2.88–4.31)     | 0.0082   |
| <i>S. officinalis</i>  | LIT      | 2.336 $\pm$ 0.39 | 0.92  | 1.29 (1.16–1.42)    | 6.52(5.28–8.03)     | 0.0092   |

\* Significant at  $p < 0.05$ **Table 5**Effect of *Lavandula angustifolia* and *Salvia officinalis* essential oil on the survival of *R. microplus* larvae.

| Essential oils         | Concentration (%) | LPT              |      |               | LIT              |      |               |
|------------------------|-------------------|------------------|------|---------------|------------------|------|---------------|
|                        |                   | Number of larvae |      | Mortality (%) | Number of larvae |      | Mortality (%) |
|                        |                   | Treated          | Dead |               | Treated          | Dead |               |
| <i>L. angustifolia</i> | 0.625             | 150              | 21   | 14.33         | 100              | 40   | 40.00         |
|                        | 1.25              | 150              | 51   | 34.44         | 100              | 66   | 66.00         |
|                        | 2.5               | 150              | 73   | 48.88         | 100              | 81   | 81.00         |
|                        | 5                 | 150              | 108  | 72.11         | 100              | 100  | 100.00        |
|                        | 10                | 150              | 125  | 83.77         | 100              | 100  | 100.00        |
|                        | Control           | 150              | 0.00 | 0.00          | 100              | 0.00 | 0.00          |
| <i>S. officinalis</i>  | 0.625             | 150              | 15   | 10.33         | 100              | 31   | 31.00         |
|                        | 1.25              | 150              | 48   | 32.55         | 100              | 45   | 45.00         |
|                        | 2.5               | 150              | 67   | 45.28         | 100              | 69   | 69.00         |
|                        | 5                 | 150              | 103  | 69.22         | 100              | 83   | 83.00         |
|                        | 10                | 150              | 121  | 81.55         | 100              | 100  | 100.00        |
|                        | Control           | 150              | 0.00 | 0.00          | 100              | 0.00 | 0.00          |

**Table 6**Effect of *Lavandula angustifolia* and *Salvia officinalis* essential oil on hatching of *R. microplus* eggs.

| Essential Oils         | Conc. (%) | No. of eggs treated | No. of eggs hatched | Hatching * (%) | Inhibition of hatching* (%) |
|------------------------|-----------|---------------------|---------------------|----------------|-----------------------------|
| <i>L. angustifolia</i> | 0.625     | 250                 | 54                  | 21.6           | 78.4                        |
|                        | 1.25      | 250                 | 45                  | 18             | 82.0                        |
|                        | 2.5       | 250                 | 38                  | 15.2           | 84.8                        |
|                        | 5         | 250                 | 0                   | 0.0            | 100                         |
|                        | 10        | 250                 | 0                   | 0.0            | 100                         |
|                        | Control   | 250                 | 246                 | 98.4           | 1.6                         |
| <i>S. officinalis</i>  | 0.625     | 250                 | 65                  | 26             | 74                          |
|                        | 1.25      | 250                 | 50                  | 20             | 80                          |
|                        | 2.5       | 250                 | 45                  | 18             | 82                          |
|                        | 5         | 250                 | 0                   | 0.0            | 100                         |
|                        | 10        | 250                 | 0                   | 0.0            | 100                         |
|                        | Control   | 250                 | 243                 | 97.2           | 2.8                         |

\* Significant at  $p < 0.01$ 

the resistance problem to several chemicals, resulting in their ineffectiveness after several years of usage (Shakya et al., 2022). Plant-derived products or bio-pesticides are convincing alternatives to conventional acaricides for tick control. Efforts are increasingly focused on developing alternatives to manufactured chemical repellents, including the extraction and assessment of plant products for repellent efficacy (Quadros et al., 2020). Plants are recognized for producing several

chemicals that serve as a defense against herbivores (Fernández-Salas et al., 2011). The use of locally sourced plant items exhibiting repellent properties is an alternate approach for the prevention of tick-borne infections.

The phytochemical analysis by GC-MS in this study showed that lavender oil is mostly made up of alpha-pinene and citrus terpenes, while sage oil is mostly made up of linalool and dihydroterpenyl which might be responsible for acaricidal activity individually or synergistically. Similarly, Kozuharova et al. (2023) identified 44 compounds in lavender essential oil containing linalyl acetate (27.5 %), linalool (24.1 %), E- $\beta$ -ocimene (7.0 %), terpinen-4-ol (5.1 %), caryophyllene (4.5 %), carvacrol (4.4 %), lavandulyl acetate (3.5 %), Z- $\beta$ -farnesene (3.3 %), and Z- $\beta$ -ocimene (3.2 %) as the major constituents. Abu-Darwish et al. (2013) conducted a GCMS analysis of sage oil that revealed a complex composition of volatile compounds in the sample having  $\alpha$ -thujone as the most abundant component (18–43 %), followed by  $\beta$ -thujone (3–8.5 %) besides other notable constituents i.e. camphor (4.5–24.5 %), 1,8-cineole (5.5–13 %), and humulene (0–12 %). Additionally, smaller quantities of  $\alpha$ -pinene (1–6.5 %), camphene (1.5–7 %), limonene (0.5–3 %), linalool (up to 1 %), and bornyl acetate (up to 2.5 %) were reported. This diverse profile underscores the aromatic and potential therapeutic properties of the plant, driven by its rich blend of terpenes and other volatile compounds. Lee et al. (2016) revealed that the chemical components of lavender oils, linalool, and linalyl acetate, exhibit superior anti-oxidative and antimicrobial activities compared to the 1, 8-cineole component. Acaricidal activity of lavender and sage

**Table 7**Repellency indices of *L. angustifolia* and *S. officinalis* essential oils tested on adult females of *R. microplus*.

| Essential oils         | Conc. (%) | No. of ticks | % of ticks attracted to |         | Repellence index ( $\pm$ SE)* | Classification** |
|------------------------|-----------|--------------|-------------------------|---------|-------------------------------|------------------|
|                        |           |              | Treatment               | Control |                               |                  |
| <i>L. angustifolia</i> | 10        | 10           | 00                      | 10      | 00 $\pm$ 00                   | Repellent        |
| <i>S. officinalis</i>  | 10        | 10           | 03                      | 7       | 0.6 $\pm$ 0.15                | Repellent        |

SE: standard error; \*\*Based on 1.0  $\pm$  SE for each treatment according to Xavier et al. (2015)

\* RI &lt; 1.0 indicates a repellent effect; RI &gt; 1.0 indicates an attractive effect; RI = 1.0 shows a neutral effect;



Fig. 1. Evaluation of repellent activity of *L. angustifolia* and *S. officinalis* essential oils against *R. microplus* larvae using trapeze test.

Table 8

Repellency activity of *Lavandula angustifolia* and *Salvia officinalis* essential oil tested on larvae of *R. microplus*.

| Essential oils         | Conc. (%) | No. of Larvae | Larvae           |             | Repellence index*<br>( $\pm$ SE) | Classification**     |
|------------------------|-----------|---------------|------------------|-------------|----------------------------------|----------------------|
|                        |           |               | Crossed the line | Not crossed |                                  |                      |
| <i>L. angustifolia</i> | 0.625     | 60            | 26               | 34          | $0.61 \pm 0.13$                  | Repellent            |
|                        | 1.25      | 60            | 24               | 36          | $0.57 \pm 0.09$                  |                      |
|                        | 2.5       | 60            | 15               | 45          | $0.40 \pm 0.11$                  |                      |
|                        | 5         | 60            | 0                | 60          | $0.0 \pm 0.0$                    |                      |
|                        | 10        | 60            | 0                | 60          | $0.0 \pm 0.0$                    |                      |
|                        | Control   | 60            | 60               | 00          | $1.0 \pm 0.0$                    |                      |
| <i>S. officinalis</i>  | 0.625     | 60            | 28               | 32          | $0.64 \pm 0.16$                  | Neutral<br>Repellent |
|                        | 1.25      | 60            | 26               | 34          | $0.61 \pm 0.13$                  |                      |
|                        | 2.5       | 60            | 16               | 44          | $0.42 \pm 0.11$                  |                      |
|                        | 5         | 60            | 8                | 52          | $0.24 \pm 0.07$                  |                      |
|                        | 10        | 60            | 0                | 60          | $0.0 \pm 0.0$                    |                      |
|                        | Control   | 60            | 60               | 00          | $1.0 \pm 0.0$                    |                      |

\*\*Based on  $1.0 \pm SE$  for each treatment according to [Xavier et al. \(2015\)](#)

\*  $RI < 1.0$  indicates a repellent effect;  $RI > 1.0$  indicates an attractive effect;  $RI = 1.0$  shows a neutral effect; SE: standard error;

Table 9

Mean ( $\pm$  SE) values of TLC (thousands / cumm), DLC, Hb (g / dL), PCV (%), TEC (millions/ cumm), AST (IU / L), ALT (IU / L) and creatinine (mg / dL) before and after treatment.

| Parameters | Groups           |                  |                  |                  |
|------------|------------------|------------------|------------------|------------------|
|            | I (Lavender EO)  |                  | II (Sage EO)     |                  |
|            | Treatment        |                  | Treatment        |                  |
|            | Pre              | Post             | Pre              | Post             |
| TLC        | $9.84 \pm 0.39$  | $8.33 \pm 0.39$  | $10.51 \pm 0.30$ | $9.32 \pm 0.12$  |
| N          | $31.66 \pm 1.76$ | $34.3 \pm 1.52$  | $30 \pm 1.18$    | $33.16 \pm 1.27$ |
| L          | $66.66 \pm 1.33$ | $65 \pm 1.36$    | $66.66 \pm 1.49$ | $65.5 \pm 1.47$  |
| M          | $0.83 \pm 0.54$  | $0.5 \pm 0.34$   | $1.50 \pm 0.5$   | $0.83 \pm 0.30$  |
| E          | $0.33 \pm 0.21$  | $00 \pm 00$      | $0.83 \pm 0.30$  | $0.16 \pm 0.16$  |
| B          | $0.5 \pm 0.34$   | $0.16 \pm 0.16$  | $1.0 \pm 0.36$   | $0.33 \pm 0.21$  |
| HB         | $8.33 \pm 0.24$  | $8.96 \pm 0.17$  | $8.30 \pm 0.22$  | $8.64 \pm 0.24$  |
| TEC        | $4.44 \pm 0.19$  | $5.02 \pm 0.31$  | $4.43 \pm 0.24$  | $4.91 \pm 0.34$  |
| PCV        | $23.08 \pm 0.80$ | $26.79 \pm 0.59$ | $24.49 \pm 1.07$ | $27.69 \pm 1.14$ |
| AST        | $93.27 \pm 5.01$ | $81.8 \pm 3.51$  | $107.6 \pm 6.19$ | $100.5 \pm 5.78$ |
| ALT        | $32.91 \pm 2.86$ | $27.5 \pm 2.17$  | $43.20 \pm 6.80$ | $37.01 \pm 6.54$ |
| Creatinine | $0.62 \pm 0.06$  | $0.51 \pm 0.04$  | $0.84 \pm 0.06$  | $0.73 \pm 0.06$  |

TLC: Total leucocyte count, N: Neutrophils, L: Lymphocytes, M: Monocytes, E: Eosinophils, B: Basophils, HB: Hemoglobin, TEC: Total Erythrocyte Count, PCV: Packed Cell Volume, AST: alanine transaminase, ALT: aspartate transaminase, Results between pre and post treatment groups were not significant at  $p < 0.05$

essential oils can be largely attributed to the presence of monoterpenes and terpenoids like linalool, thujone, camphor, and 1,8-cineole which act through neurotoxic mechanisms, either disrupting neural signaling pathways or causing metabolic dysfunction in ticks. The literature supports their individual and synergistic efficacy, providing a strong basis for considering these EOs as natural acaricides ([Stefanidesova et al., 2017](#)).

[Olivia Chavez et al. \(2023\)](#) evaluated the impact of five organic compounds including lavender oil on *Rhipicephalus (Boophilus) microplus* larvae through LIT and reported 90 % mortality at 10 % concentration which supports the present findings where hundred percent larvae were killed with 5 and 10 % lavender oil. [Shalaby et al. \(2016\)](#) discovered that essential oil from *L. officinalis* and camphor at a 32 % concentration had significant larvicidal efficacy against the sheep blowfly, *Lucilia serrata* causing 100 % and 93.33 % larval mortality, respectively.

The current investigation found no hatching of *R. microplus* eggs treated with 5 and 10 % doses of *L. angustifolia* and *S. officinalis* essential oils. Likewise, [Pirali-Kheirabadia and da Silva et al. \(2010\)](#) examined acaricidal properties of *L. angustifolia* essential oil on engorged female *R. (B.) annulatus* showing a dose-dependent decrease in egg weight, with complete suppression of egg laying at 6 and 8 %. [Singh et al. \(2022\)](#) assessed the anti-tick effectiveness of eight essential oils (EOs) including *S. officinalis* against *R. microplus* and documented more than 95 % effectiveness on survival and reproduction potential with this EO.

The current research indicates that both essential oils have repellent properties against adult and larval stages of *R. microplus*. Similarly,



**Fig. 2.** In vivo trials of *L. angustifolia* and *S. officinalis* essential oils. Figs. a & B: tick infestation on cattle, fig. C: engorged ticks dropped on the ground, figs. D & E: application of essential oils on cattle by spraying method.

**Table 10**

In vivo efficacy of *L. angustifolia* and *S. officinalis* essential oils (at 5 % dose) in tick infested cross breed cattle.

| Essential oils         | No. of ticks survived on the animals (hours) |        |        |        | Efficacy (%)<br>after 72 h |
|------------------------|----------------------------------------------|--------|--------|--------|----------------------------|
|                        | 0                                            | 24     | 48     | 72     |                            |
| <i>L. angustifolia</i> | 27.83                                        | 21.00  | 10.33  | 3.16   | 88.64                      |
|                        | ± 2.16                                       | ± 1.50 | ± 0.76 | ± 0.30 |                            |
| <i>S. officinalis</i>  | 25.00                                        | 18.83  | 12.16  | 3.83   | 84.68                      |
|                        | ± 1.30                                       | ± 1.24 | ± 1.49 | ± 0.60 |                            |

Mkolo and Magano (2007) reported 70–100 % repellent activity of 5, 10, and 20 % *L. angustifolia* essential oil against adult *Hyalomma marginatum rufipes* under climbing test. Jaenson et al. (2006) investigated the effects of lavender and geranium oils on *Ixodes ricinus* nymphs and reported that 1 % lavender oil attracted 45 % of nymphs after 5 min, but the control group attracted 92 % while 30 % lavender oil repelled all the tested nymphs.

Jain et al. (2021) reported a significant reduction in tick counts after 144 hrs under in vivo trial of *Citrus limetta* seed oil (CLO) against *R. microplus* infested cattle and observed clinical safety of this oil exhibiting no signs of skin irritation and normal hematological parameters. Similarly, in our in vivo trial, both the essential oils provided more than 84 % efficacy after 72 h and showed no adverse skin effects. Gonzaga et al. (2023) identified *R. microplus* as the most extensively researched tick species and the majority of studies (74 %) employed

topical applications of essential oils. In this way, these EOs can be employed for the development of eco-friendly herbal solutions for managing tick problems in cattle.

## 5. Conclusion

The present findings revealed that these oils have various anti-tick properties against different tick stages which can provide a complete solution for tick management. Also, these EOs or their potential constituents can be exploited to develop a commercial formulation for a complete package of tick eradication from animals that can minimize the frequency of chemical application on animals in an eco-friendly manner.

## Ethical standards and consent to participate

The present study was approved by the Institutional Ethical Committee of the College of Veterinary Science & Animal Husbandry (NDVSU), Mhow, Madhya Pradesh, India.

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## CRediT authorship contribution statement

**Mamta Singh:** Writing – original draft, Formal analysis. **R.K. Bagherwal:** Supervision, Resources, Funding acquisition. **Anil Kumar Sharma:** Writing – review & editing, Writing – original draft. **Hemant Kumar Mehta:** Writing – review & editing. **Mukesh Shaky:** Writing – original draft, Resources, Funding acquisition, Formal analysis, Conceptualization. **Shivam Singh Mehrotra:** Validation, Methodology, Investigation.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Availability of data and materials

The derived data supporting this study's findings are available from the corresponding author upon request.

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